

Question Paper

Exam Date & Time: 30-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Reinforcement Learning [NPT_4126]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the exploration-exploitation tradeoff in RL? Give a short example. (2)
- A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Describe the multi-armed bandit problem and explain how it represents the exploration-exploitation dilemma. (3)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- C) Interpret how the Policy Gradient Theorem helps in directly optimizing stochastic policies. Use an example of a simple environment (e.g., two-action system) to show how gradients are estimated and used to update the policy parameters. Discuss the advantages of this approach compared to value-based methods. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Discuss the PAC (Probably Approximately Correct) framework in the context of reinforcement learning. (2)
- A) Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) Interpret the Policy Gradient Theorem and discuss how it helps in optimizing policy directly. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Compare PAC-based evaluation with regret-based evaluation methods in reinforcement learning. Highlight key differences in their goals and interpretations. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Distinguish between value-based and policy-based reinforcement learning methods. (2)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Analyze how the Bellman Optimality Principle helps in determining the optimal policy in reinforcement learning problems. (3)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- C) For an autonomous delivery robot navigating between three locations, define an appropriate (5)

Markov Decision Process (MDP) including states, actions, transition probabilities, and rewards. Using this setup, apply the Bellman equation for the state-value function to compute one iteration of value updates for each state.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4) Apply the SARSA and Q-learning algorithms on a simple grid-world problem and illustrate how the Q-values are updated after one episode. (3)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Demonstrate with an example how n-step returns differ from 1-step Temporal Difference (TD) learning. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Interpret the performance of Deep Q-Networks (DQN) compared to traditional Q-learning, citing the role of experience replay and target networks. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Interpret how the Options Framework in Hierarchical Reinforcement Learning (HRL) helps an agent solve a multi-step task such as "navigating a maze with sub-goals." (3)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Examine the role of temporal abstraction in Hierarchical RL, and how it improves learning efficiency in complex tasks. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Given a scenario of a self-driving car operating under uncertain sensor inputs, apply the POMDP framework to model the problem. Describe possible solution methods for policy optimization under partial observability. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4126:Reinforcement Learning

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	15
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4126:Reinforcement Learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)

2	Understand	3	7(14%)
3	Application	8	30(60%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)